

Robot Academy — Feature Register

Current version: M1.1 · Playable Slice, audited

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This is the living feature register. It is regenerated into docs/FEATURES.pdf at every milestone, so the PDF always describes the current state of the build. Status values: SHIPPED, PARTIAL, PLANNED.

Milestone M1 — Playable Slice

Engine and rendering

ID	FEATURE	STATUS	FILES
E-01	WebGL renderer with soft shadow mapping and device-pixel-ratio clamping	SHIPPED	src/core/engine.js
E-02	Orbiting follow camera — fixed pitch, yaw on right-drag, scroll to zoom	SHIPPED	src/core/engine.js, src/core/input.js
E-03	Multi-scene manager — overworld and interiors are separate scenes, swapped on entry	SHIPPED	src/core/engine.js, src/main.js
E-04	Shared lighting rig with separate indoor and outdoor profiles	SHIPPED	src/core/engine.js
E-05	Flat-shaded material helper — the chunky low-poly look is enforced in one place	SHIPPED	src/core/engine.js
E-06	Frame loop with delta clamping so tab-switching doesn't teleport the player	SHIPPED	src/core/engine.js

World

ID	FEATURE	STATUS	FILES
W-01	Circular terrain with per-vertex colour variation (no flat green slab)	SHIPPED	src/world/village.js
W-02	Central plaza with a six-slab monument, one colour per prompt slot	SHIPPED	src/world/village.js
W-03	Seven buildings in a ring, each with pitched roof, door, windows, chimney	SHIPPED	src/world/village.js
W-04	Radial paths from the plaza to every building	SHIPPED	src/world/village.js
W-05	Canvas-texture hanging signboards — real text in 3D with no font loader	SHIPPED	src/world/village.js
W-06	Deterministic prop scatter (trees, rocks) — identical village every load	SHIPPED	src/world/village.js
W-07	Boundary tree ring so the world reads as enclosed	SHIPPED	src/world/village.js
W-08	Glowing door pads marking every enterable building	SHIPPED	src/world/village.js

Characters

ID	FEATURE	STATUS	FILES
C-01	Procedural robot builder — seven body archetypes, zero external assets	SHIPPED	src/entities/robot.js
C-02	Per-robot character props: clipboard, magnifying lens, crate stack, drones, battle damage	SHIPPED	src/entities/robot.js
C-03	Idle and walk animation — arm and leg swing, head sway, antenna pulse	SHIPPED	src/entities/robot.js
C-04	Character-specific idle behaviour (three heads bicker, crates teeter, drones orbit)	SHIPPED	src/entities/robot.js
C-05	Floating name tags that fade in by proximity	SHIPPED	src/world/village.js, src/main.js
C-06	Player robot with distinct white-and-mint palette	SHIPPED	src/entities/player.js

Movement and interaction

ID	FEATURE	STATUS	FILES
M-01	Click-to-move — raycast the ground, walk there, RuneScape style	SHIPPED	src/core/input.js, src/entities/player.js
M-02	Steering obstacle avoidance — slide around houses instead of stopping dead	SHIPPED	src/entities/player.js
M-03	Click a robot or door to walk over and act automatically on arrival	SHIPPED	src/main.js
M-04	Smoothed turn-to-face-travel-direction	SHIPPED	src/entities/player.js
M-05	Animated click marker ring	SHIPPED	src/main.js
M-06	Hover highlighting with a contextual action prompt	SHIPPED	src/main.js, src/ui/hud.js
M-07	World boundary clamping	SHIPPED	src/entities/player.js
M-08	Drag-versus-click discrimination so camera orbit never triggers a move	SHIPPED	src/core/input.js

Interiors

ID	FEATURE	STATUS	FILES
I-01	Enterable houses — walk onto the door pad and the interior scene loads	SHIPPED	src/world/interior.js, src/main.js
I-02	Interiors built lazily and cached, so re-entry is instant	SHIPPED	src/main.js
I-03	Room shell keyed to each robot's palette (rug, skirting, wall caps)	SHIPPED	src/world/interior.js
I-04	Six themed furniture sets — no two rooms look alike	SHIPPED	src/world/interior.js
I-05	Fixed dollhouse camera indoors, framing the whole room	SHIPPED	src/main.js
I-06	Exit pad returns you to the doorway you came in through	SHIPPED	src/main.js
I-07	Warm point-light lamp per room	SHIPPED	src/world/interior.js

Dialogue and curriculum

ID	FEATURE	STATUS	FILES
D-01	Typewriter dialogue overlay, click or space to advance, click again to skip the reveal	SHIPPED	src/systems/dialogue.js
D-02	Paged lessons with section headings and a page counter	SHIPPED	src/systems/dialogue.js
D-03	Speaker accent colour drawn from the robot's own palette	SHIPPED	src/systems/dialogue.js
D-04	Context-aware dialogue — outdoor greeting versus the full indoor lesson	SHIPPED	src/main.js
D-05	Repeat visits open with different framing	SHIPPED	src/main.js
D-06	Six full curricula, 36 lesson pages, adapted from The Prompting Field Manual	SHIPPED	src/data/robots.js
D-07	Keyboard control — Space/Enter advance, Escape closes	SHIPPED	src/systems/dialogue.js

The Forge (prompt builder)

ID	FEATURE	STATUS	FILES
F-00	The Forge is a real room you walk into — six lit pedestals, one per slot, and a terminal you operate	SHIPPED	src/world/interior.js
F-01	Six-slot prompt builder matching the manual's anatomy	SHIPPED	src/ui/forge.js
F-02	Live assembly — the brief rebuilds as you type	SHIPPED	src/ui/forge.js
F-03	Per-slot hint text explaining what that slot is for	SHIPPED	src/ui/forge.js
F-04	Worked placeholder examples in every field	SHIPPED	src/ui/forge.js
F-05	Posture selector — interview me / plan first / act autonomously / give options	SHIPPED	src/ui/forge.js
F-06	Completeness meter with diagnostic feedback naming the missing slot	SHIPPED	src/ui/forge.js
F-07	Copy to clipboard with a execCommand fallback	SHIPPED	src/ui/forge.js

HUD and progression

ID	FEATURE	STATUS	FILES
H-01	Lesson progress bar, 0–6	SHIPPED	src/ui/hud.js
H-02	Journal panel listing every robot, its subject, and what you earned	SHIPPED	src/ui/hud.js
H-03	Progress persisted to localStorage, wrapped for private-browsing failure	SHIPPED	src/ui/hud.js
H-04	Toast notifications for lessons learned and room entry	SHIPPED	src/ui/hud.js
H-05	Persistent control hints	SHIPPED	src/ui/hud.js
H-06	Completion message when all six lessons are done	SHIPPED	src/main.js

Build and delivery

ID	FEATURE	STATUS	FILES
B-01	No build step — ES modules with a CDN importmap, runs from any static server	SHIPPED	index.html
B-02	One concern per module, so git diffs stay readable	SHIPPED	src/**
B-03	Responsive UI down to phone width	SHIPPED	styles/ui.css
B-04	prefers-reduced-motion respected	SHIPPED	styles/ui.css
B-05	Headless WebGL test harness (render, enter house, open dialogue)	SHIPPED	tools/test-game.mjs
B-06	Loading screen removed only once the scene is live	SHIPPED	index.html, src/main.js
B-07	Full functional audit — 42 automated checks over every interaction path	SHIPPED	tools/audit.mjs
B-08	Debug/automation seam (window.RA.debug) so tests drive the game directly	SHIPPED	src/main.js
B-09	Inline SVG favicon — no 404 on load	SHIPPED	index.html

Audit — M1.1

`node tools/audit.mjs` boots the game headlessly and runs **42 checks** across boot, content integrity, movement, all six robots, all six interiors, every lesson, the Forge, progression, persistence, HUD, camera and responsive layout.

Result: 42/42 passing, stable across three consecutive runs.

Four defects were found and fixed during the audit:

ID	DEFECT	FIX
A-01	Clicking a robot standing near its own doorway entered the house instead of talking — the flat door pad sat nearer the camera along the view ray, so nearest-hit-wins picked the wrong thing	Intent-ranked picking: collect every ray hit and prefer robot over terminal over exit over door
A-02	Walking to a point inside a building ground against the obstacle forever, the player never stopping	<code>moveTo()</code> clamps any destination inside a blocker out to that building's edge, plus a stall timer as a safety net
A-03	The Forge opened its panel without you ever entering the building, contradicting the brief	Built The Forge as a real interior — six lit pedestals and a terminal you walk up to and click
A-04	A 404 on every page load (missing favicon)	Inline SVG favicon

Dead code removed in the same pass: an unused `nearest()` helper, an unused `hovered` variable, an unused `sky` import, and an animation line reading properties that were never set.

Known limitations in M1.1

- **Collision is steering-based, not solid.** You slide around buildings rather than being hard-blocked. Walking into a wall at an angle can clip a corner.
- **No pathfinding.** The player walks in a straight line toward the click and pushes off obstacles. It cannot route around a building to reach a point behind it — you click twice instead.

- **Interior furniture is randomised per session**, not seeded like the village, so book and crate positions differ between reloads.
- **Draw calls are unbatched** (~660 in the overworld). Fine on desktop, worth merging static geometry before targeting low-end mobile.
- **Mobile is untested**. Touch produces pointer events so it should work, but the camera orbit has no touch gesture and the HUD is laid out for desktop.
- **No audio**.

Planned — not yet built

ID	FEATURE	MILESTONE
P-01	XP, levels and a skill panel (Prompting / Context / Agents)	M2
P-02	Quest log framing each robot as a completable quest	M2
P-03	Per-robot mini-game testing the lesson just taught	M3
P-04	Forge presets seeded by which robots you have learned from	M2
P-05	Save/load of forged prompts	M2
P-06	Audio — footsteps, ambient birds, dialogue blips	M3
P-07	Mobile touch controls and layout	M3
P-08	Geometry merging for draw-call reduction	M3
P-09	Day/night cycle	Backlog
P-10	A seventh robot covering evals and measurement	Backlog

Decision log

Decisions taken with the project owner rather than assumed.

DATE	DECISION	CHOSEN	WHY
2026-09-12	Version control route	Local folder + GitHub Desktop	Claude cannot run git on the owner's machine (September Windows update broke the workspace mount); GitHub Desktop supplies the git engine without a token passing through chat
2026-09-12	Project structure	Modular ES modules, no build step	File-per-feature git diffs, no <code>npm install</code> , deploys to GitHub Pages as-is
2026-09-12	Movement	Click-to-move	Faithful to the RuneScape reference, works on touch, keeps the camera simple
2026-09-12	M1 scope	World, robots, dialogue, interiors, Forge — no XP yet	Ship a tight vertical slice and expand, rather than half-build six systems

DATE	DECISION	CHOSEN	WHY
2026-09-12	Curriculum source	The Prompting Field Manual	Content already exists and is sourced; build time goes into the game
2026-09-12	Art direction	Chunky low-poly built in code	No external assets, fully version-controlled, distinctive rather than generic
2026-09-12	Hosting target	GitHub Pages, public, desktop-first	Free, shareable, straight from the repo
2026-09-12	Feature documentation	Living doc regenerated per milestone	Always current; markdown source stays diffable in git
2026-09-12	Verification approach	Automated audit harness over manual play-testing	42 checks re-runnable on every change; caught four defects manual play would likely have missed